S-CLIP: Semi-supervised Vision-Language Learning using Few Specialist Captions



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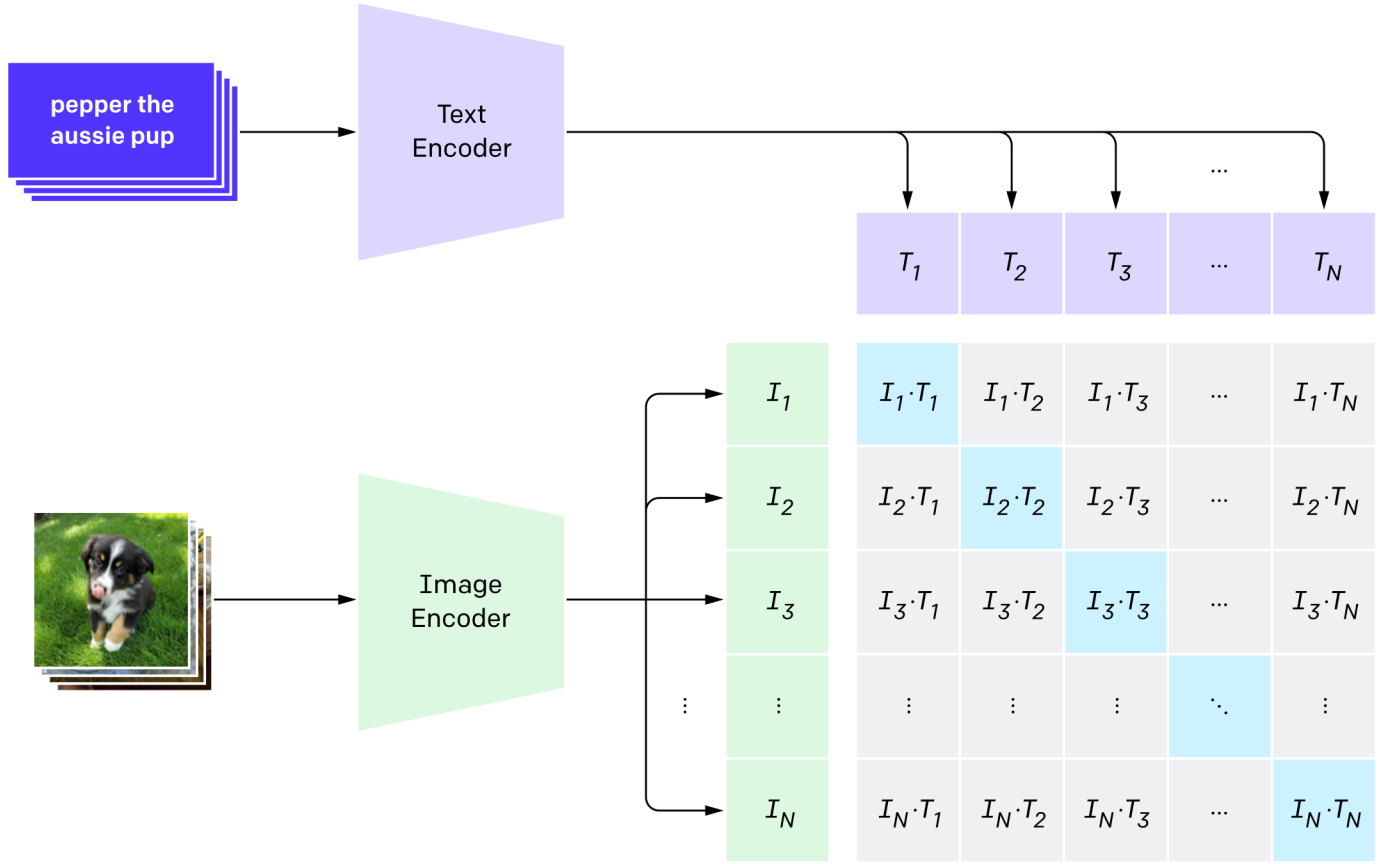
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KAIST

Motivation: CLIP for Specialist Domains

CLIP is a vision-language model that **connects image & text** embeddings



Learning Transferable Visual Models From Natural Language Supervision. ICML'21.

Motivation: CLIP for Specialist Domains

However, CLIP often fails in **specialist domains** where captions are hard to obtain



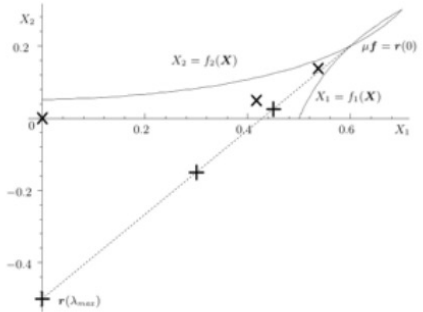
"many houses arranged neatly and some roads crossed them with some plants in the roadside"

(a) Remote sensing



"grey classic u-neck t-shirt. short sleeve t-shirt in heather grey. tonal logo flag at u-neck collar."

(b) Fashion



"Fig. 2. variable 2-D initial probe distribution used for CFO runs reported here"

(c) Scientific figure



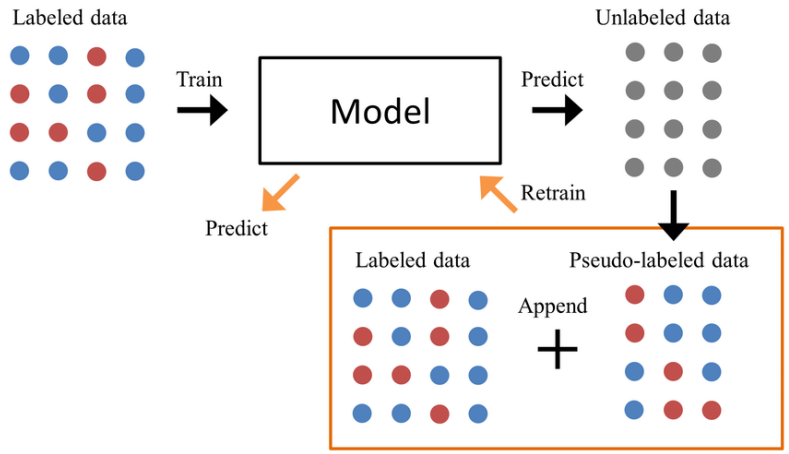
"simpsons wearing headphones and looking like a guy"

(d) Comics

→ *Semi-supervised (**Semi-SL**) setup with a few labeled image-text pairs and many unlabeled images*

Challenge: Semi-SL fails for CLIP

Semi-SL = Assign **pseudo-labels (PL)** to unlabeled images and retrain the model with them



However, **naïve PL fails** for CLIP because each image has a different caption, unlike discrete classes



"two tennis courts and some buildings are in two sides of a road"

Nearest image in the batch

Their captions are different! (naive pseudo-labeling fails)

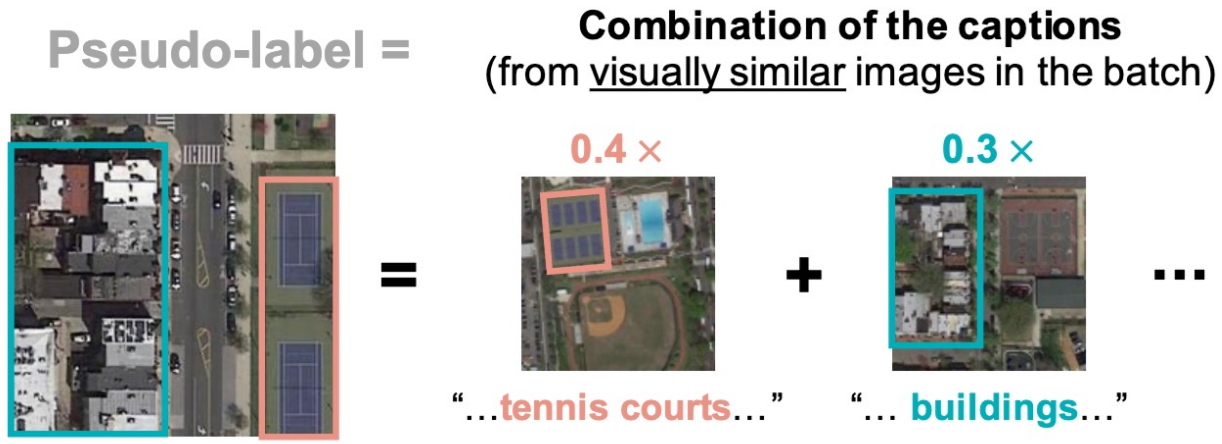


"the huge rectangle contains a baseball field, a swimming pool, two basketball courts and eight tennis courts"

Our Method: S-CLIP

1. Caption-level PL:

Semantic of an unlabeled image is a combination of the captions of visually similar images

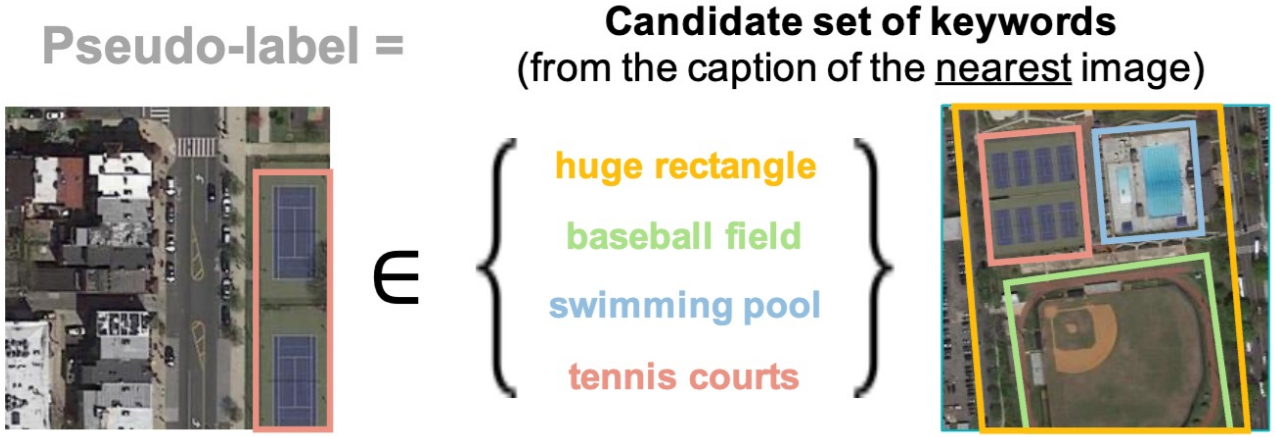


→ *Solve an optimal transport (OT) problem*

Our Method: S-CLIP

2. Keyword-level PL:

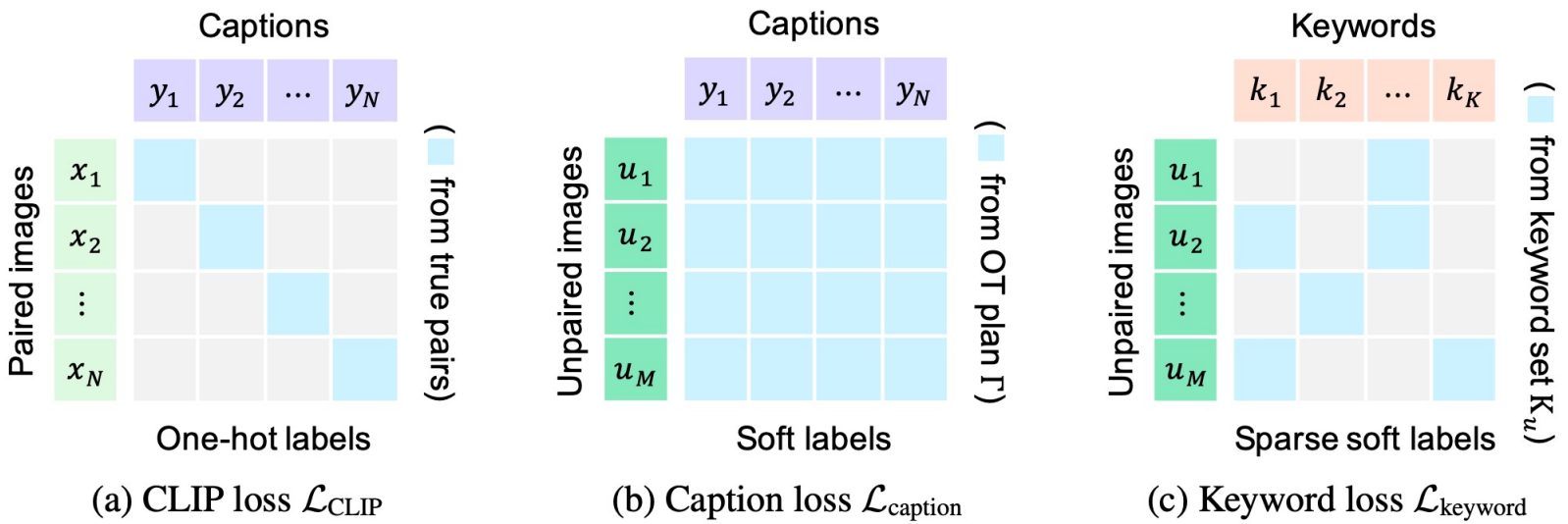
Unlabeled image shares some keywords with the nearest labeled image (although not all of them match)



→ *Solve a partial label learning (PLL) problem*

Our Method: S-CLIP

Training objective: Soft version of the CLIP loss



(a) CLIP loss $\mathcal{L}_{\text{CLIP}}$ (b) Caption loss $\mathcal{L}_{\text{caption}}$ (c) Keyword loss $\mathcal{L}_{\text{keyword}}$

Benefits of S-CLIP

Performance: Better zero-shot classification & image-text retrieval than CLIP

Method	Data	RSICD-CLS	UCM-CLS	WHU-RS19	RSSCN7	AID
CLIP (original)	-	45.3	50.5	65.5	58.9	47.8
CLIP (fine-tune)	L	58.3±0.3	63.5±3.4	76.5±3.2	61.9±1.2	63.1±1.3
Hard-PL [16]	L=U	56.6±3.5 (-1.7)	61.6±2.2 (-1.9)	78.1±2.5 (+1.6)	63.9±2.1 (+2.0)	63.2±2.6 (+0.1)
Soft-PL [21]	L=U	62.5±0.8 (+4.2)	65.7±2.7 (+2.2)	83.7±2.7 (+7.2)	65.7±0.6 (+3.8)	68.0±0.7 (+4.9)
S-CLIP (ours)	L=U	**66.9**±1.7 (+8.6)	**66.7**±1.6 (+3.2)	**86.9**±2.0 (+10.4)	**66.2**±1.1 (+4.3)	**73.0**±0.3 (+9.9)

		Image→text R@5			Text→image R@5		
Method	Data	RSICD	UCM	Sydney	RSICD	UCM	Sydney
CLIP (original)	-	9.4	34.3	36.2	10.1	24.8	51.7
CLIP (fine-tune)	L	15.4±1.7	41.3±1.8	47.1±6.5	15.1±1.0	40.9±1.6	56.1±2.4
Hard-PL [16]	L=U	16.1±0.2	40.8±2.9	43.1±3.0	15.7±0.7	40.5±3.0	47.7±5.3
Soft-PL [21]	L=U	17.0±0.9	43.2±3.9	42.0±4.3	16.5±0.1	42.9±3.3	50.2±4.9
S-CLIP (ours)	L=U	**18.4**±0.6	**45.7**±1.4	**50.0**±3.0	**16.8**±1.2	**43.5**±1.5	**55.1**±2.0

Benefits of S-CLIP

Robust to domain shifts: In practice, labeled & unlabeled data can be different

Labeled

(e.g., data from San Francisco)



≠

Unlabeled

(e.g., data from Seoul)



Method	Data	RSICD-CLS	UCM-CLS	WHU-RS19	RSSCN7	AID
CLIP (original)	-	45.3	50.5	65.5	58.9	47.8
CLIP (fine-tune)	L	58.3±0.3	63.5±3.4	76.5±3.2	61.9±1.2	63.1±1.3
Hard-PL [16]	L≠U	55.8±1.2 (-2.5)	61.0±6.2 (-2.5)	76.3±1.1 (-0.2)	62.5±2.7 (+0.6)	62.3±1.4 (-0.8)
Soft-PL [21]	L≠U	56.1±3.5 (-2.2)	62.4±1.8 (-1.1)	79.5±3.6 (+3.0)	63.3±2.6 (+1.4)	62.4±2.1 (-0.7)
S-CLIP (ours)	L≠U	**65.3**±2.1 (+7.0)	**66.0**±0.4 (+2.5)	**86.3**±1.1 (+9.8)	**68.3**±2.2 (+6.4)	**70.8**±2.5 (+7.7)